

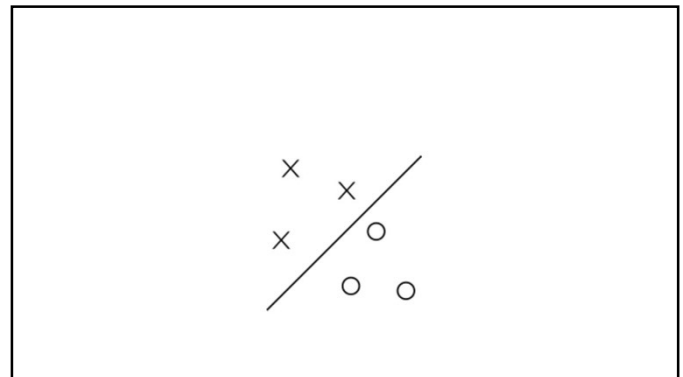


STROJNO UČENJE

8. predavanje

Još klasifikacijskih metoda

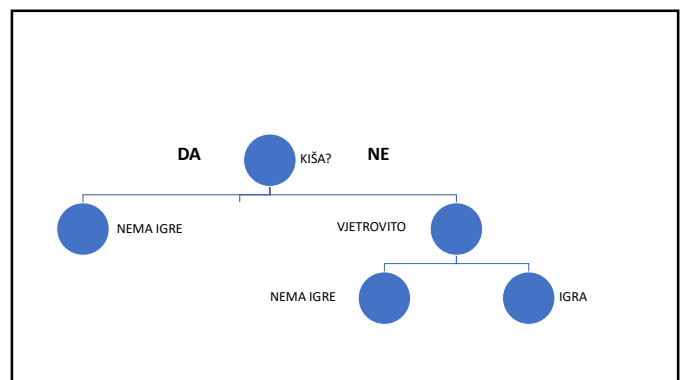
AI Center Lipik
THE FUTURE IS HERE



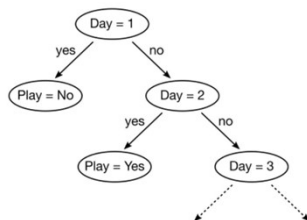
8.1. Stabla odlučivanja



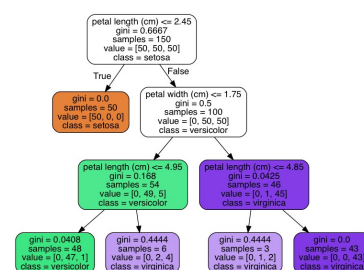
OBLAČNO	KIŠA	VIETROVIT O	TEMPERAT URA	IGRANJE
DA	DA	DA	23	NE
NE	NE	NE	26	DA
DA	NE	NE	25	DA
DA	NE	NE	15	DA
NE	NE	DA	16	NE

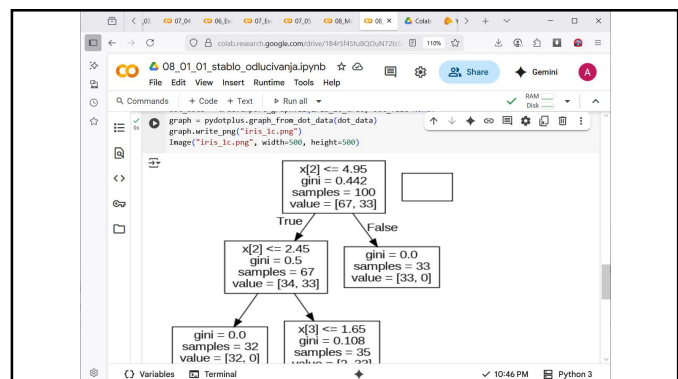
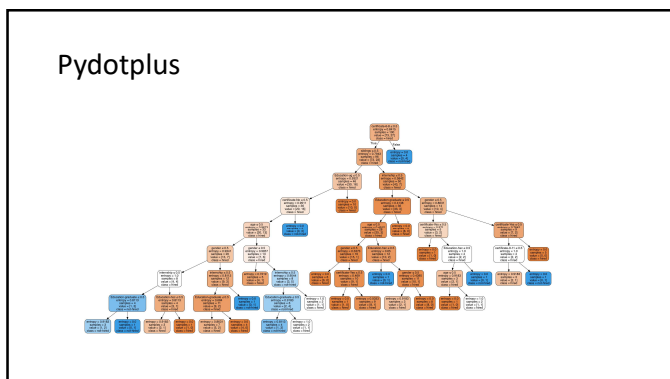
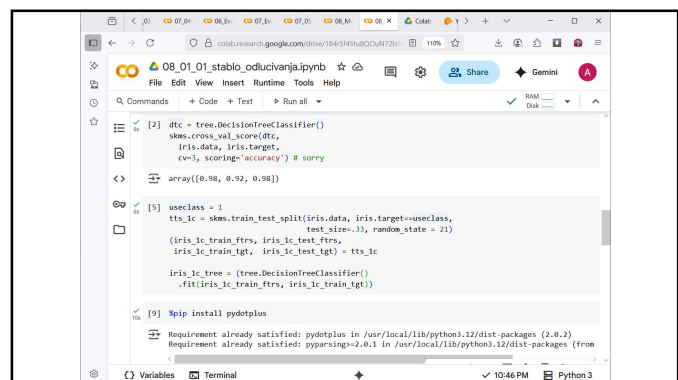
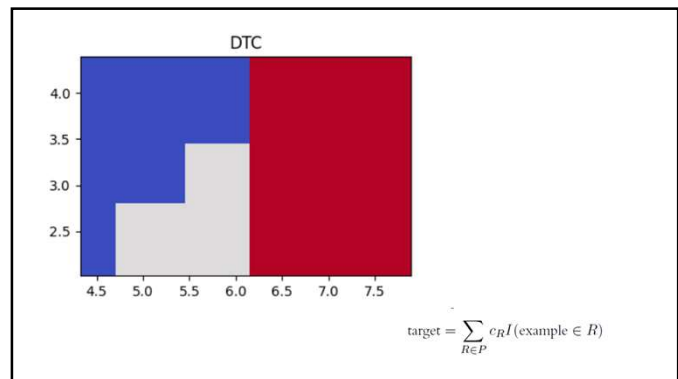
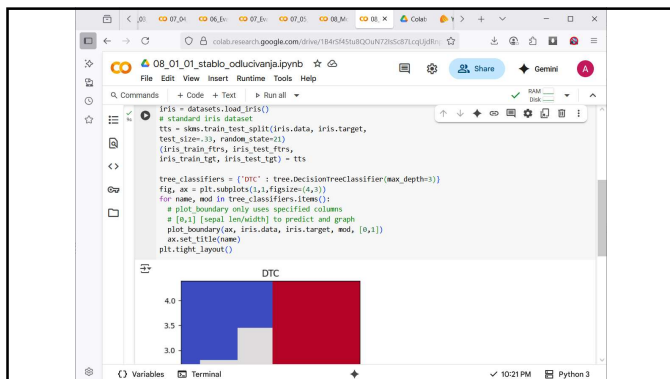


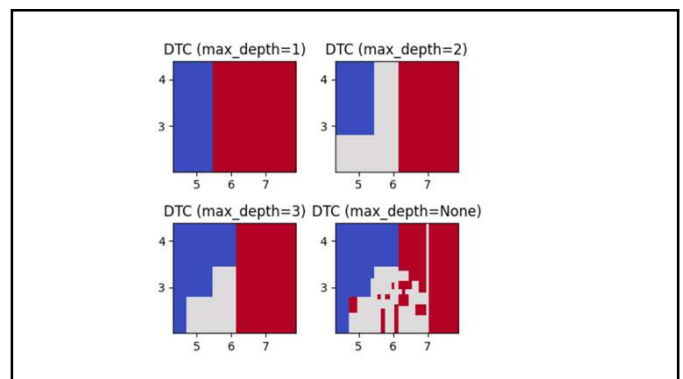
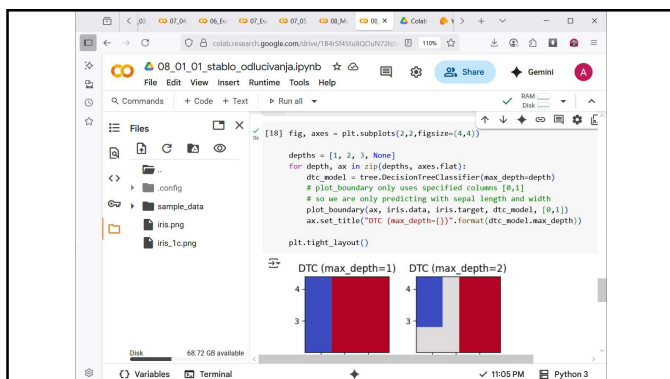
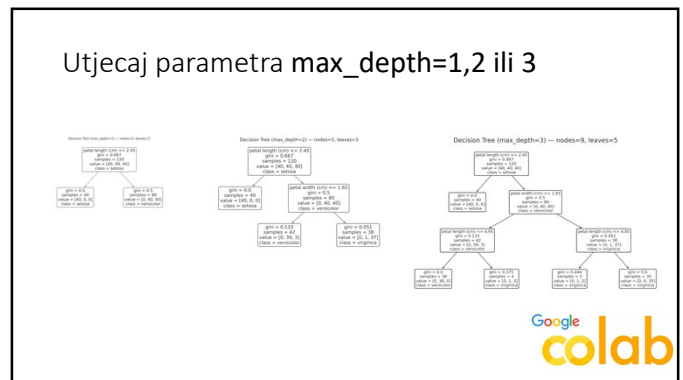
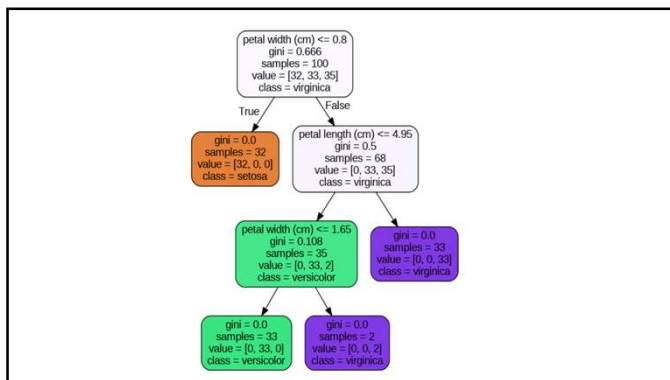
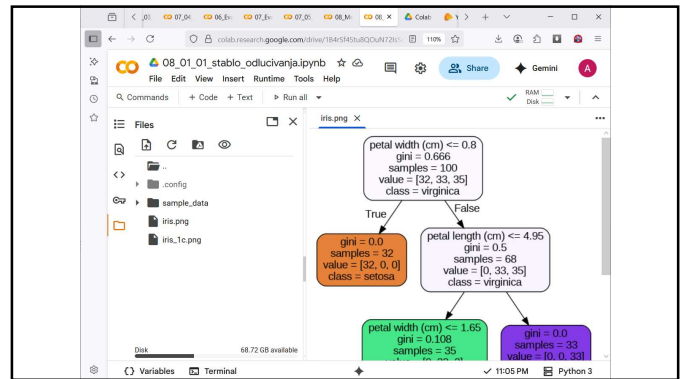
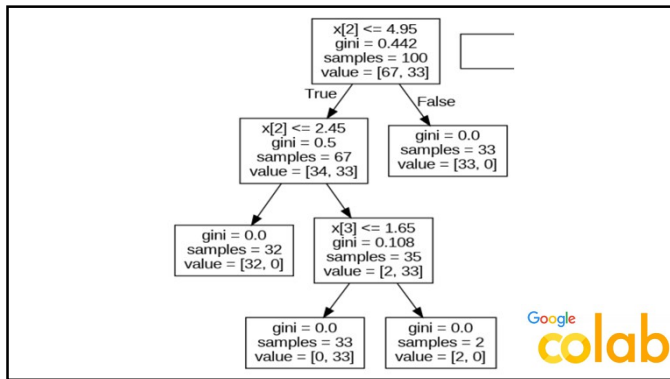
Problem jedinstvenih identifikatora



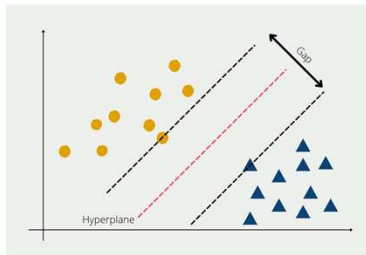
<https://gist.github.com/kdokic1971/f74dcafdb4f49bdcdb9992354856724>







8.2 Klasifikatori potpornih vektora



The screenshot shows a Google Colab notebook with the following code:

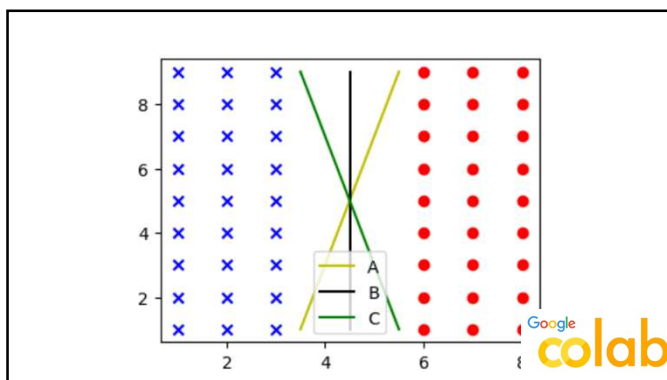
```
import numpy as np
import matplotlib.pyplot as plt

fig, ax = plt.subplots(1,1,figsize=(4,3))

# fancy way to get cross-product of points
left = np.mgrid[1:4,0, 1:10].reshape(2, -1).T
right = np.mgrid[6:9,0, 1:10].reshape(2, -1).T

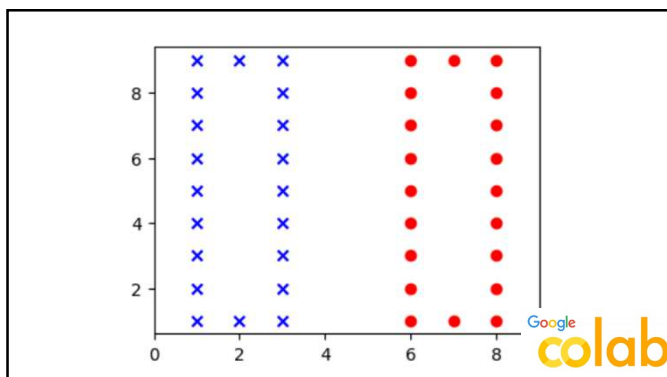
# data points
ax.scatter(left[:,0], left[:,1], c='b', marker='x')
ax.scatter(right[:,0], right[:,1], c='r', marker='o')

# separating lines
ax.plot([3.5, 5.5], [1,9], 'y', label='A')
ax.plot([4.5, 4.5], [1,9], 'k', label='B')
ax.plot([3.5, 5.5], [9,1], 'g', label='C')
ax.legend(loc='lower center')
```



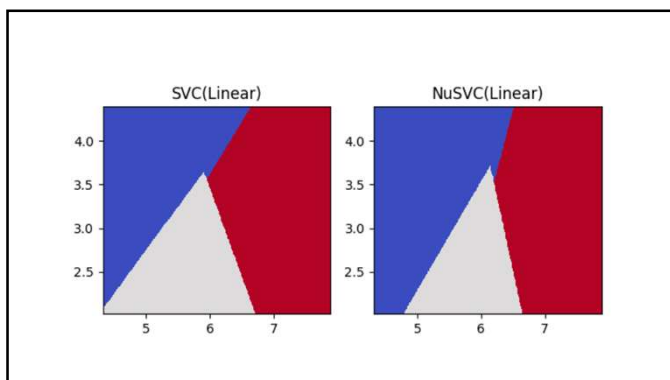
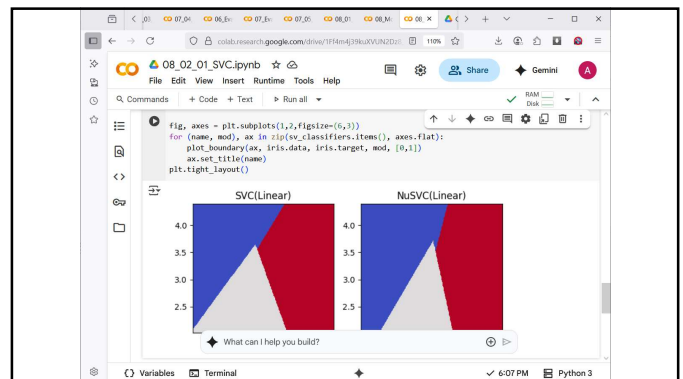
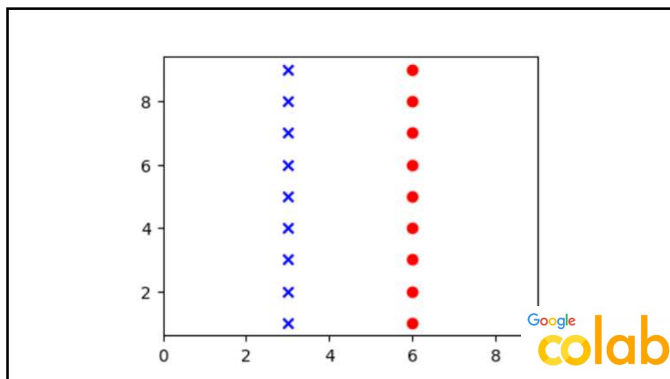
The screenshot shows the same code as the previous notebook, but with an additional line to set the x-axis limits:

```
ax.set_xlim(0,9)
```



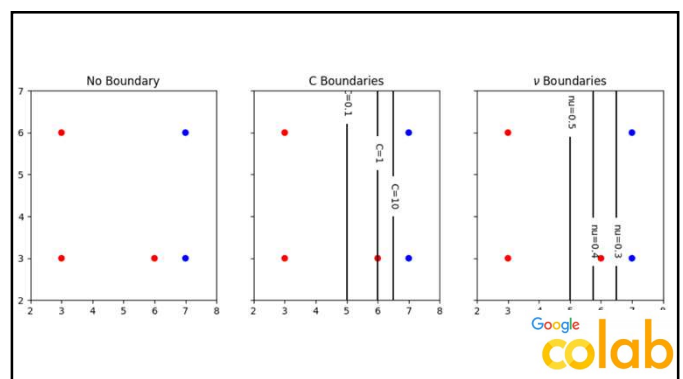
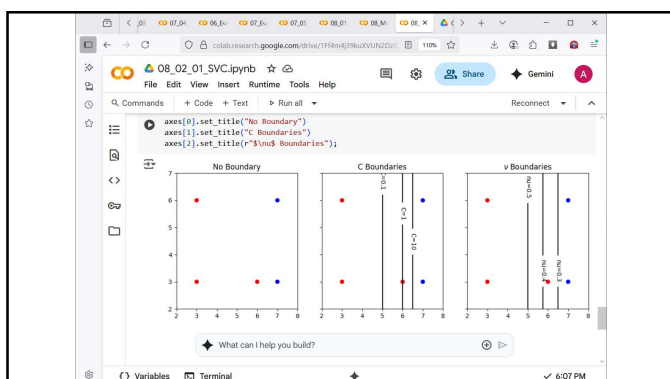
The screenshot shows the same code as the previous notebook, but with an additional line to set the x-axis limits:

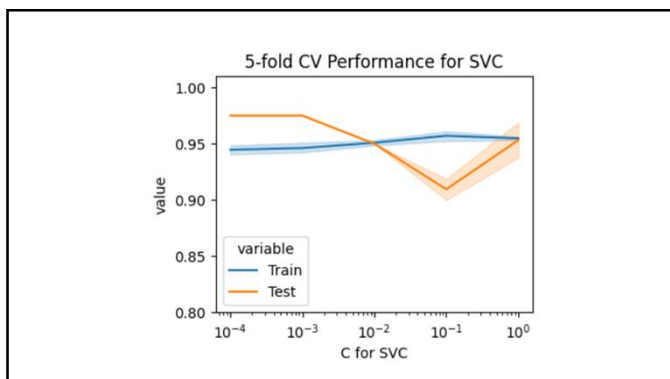
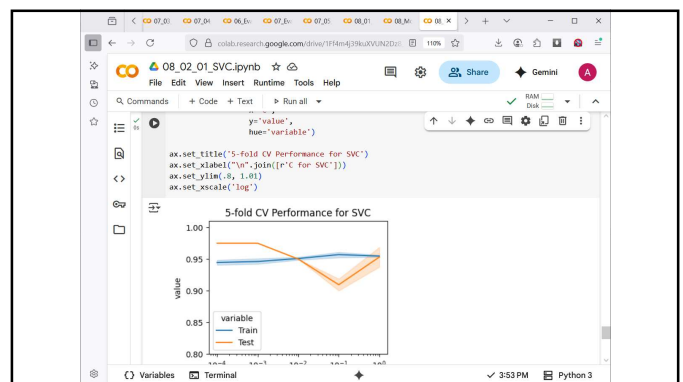
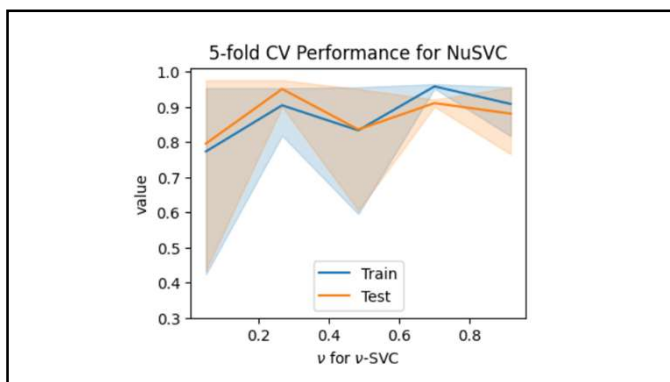
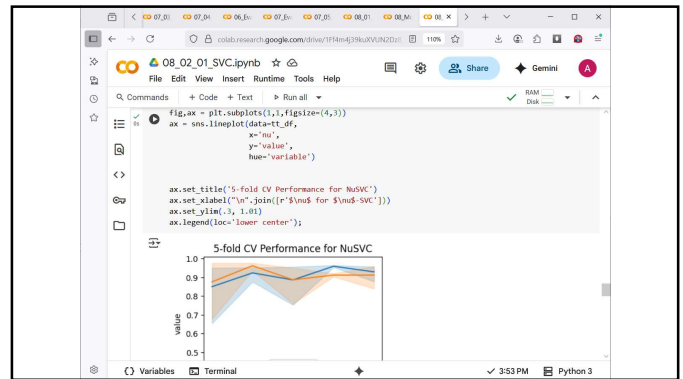
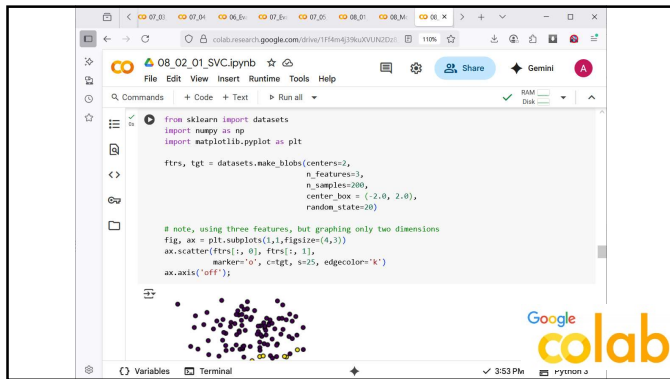
```
ax.set_xlim(0,9)
```



8.2.1. Parametri za SVC

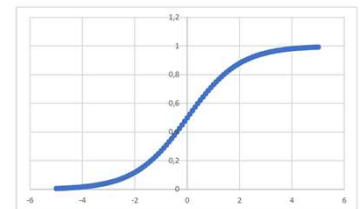
- SVC – parametar C
- NuSVC – parametar ν (najmanje $\nu\%$ podataka bit će zadržano kao vektor podrške)





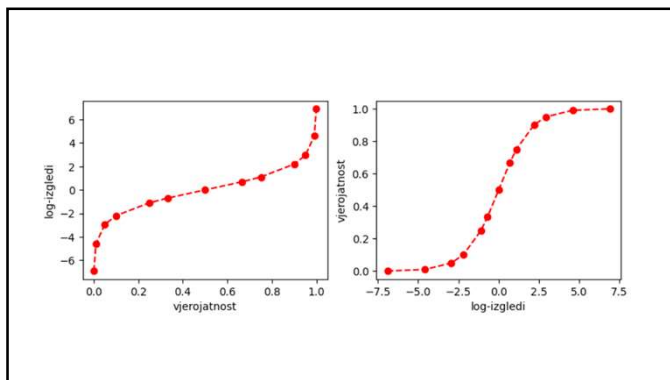
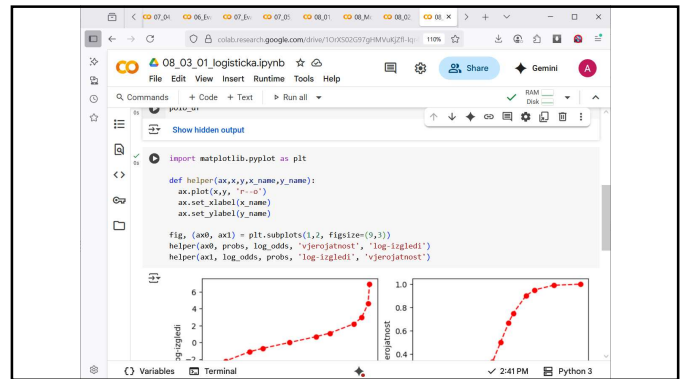
8.3. Logistička regresija

$$y = \frac{1}{1 + e^{-x}}$$



$$y = \frac{e^{\beta_0 + x \cdot \beta_1}}{1 + e^{\beta_0 + x \cdot \beta_1}}$$

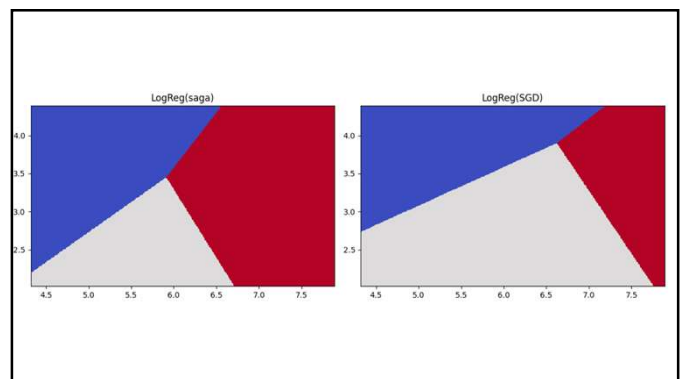
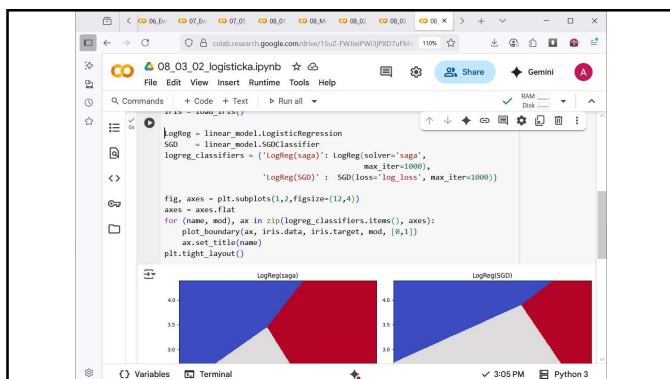
Google
colab



Metode

- SGDClassifier
- LogisticRegression

Google
colab



8.4. Diskriminantna analiza

- **Kovarijanca** je statistička mjera koja pokazuje kako se dvije varijable mijenjaju zajedno.

- Ako se dvije varijable povećavaju ili smanjuju zajedno → pozitivna kovarijanca
- Ako jedna raste dok druga pada → negativna kovarijanca
- Ako nema uzajamne veze → kovarijanca je blizu nule

Za dvije varijable X i Y , kovarijanca je:

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

gdje su:

- x_i, y_i : i -ti par podataka
- $\bar{x}, \bar{y}$: srednje vrijednosti X i Y
- n : broj uzoraka

X (visina)	Y (masa)
160	55
170	65
180	75

Kad X raste, Y također raste → kovarijanca je pozitivna.

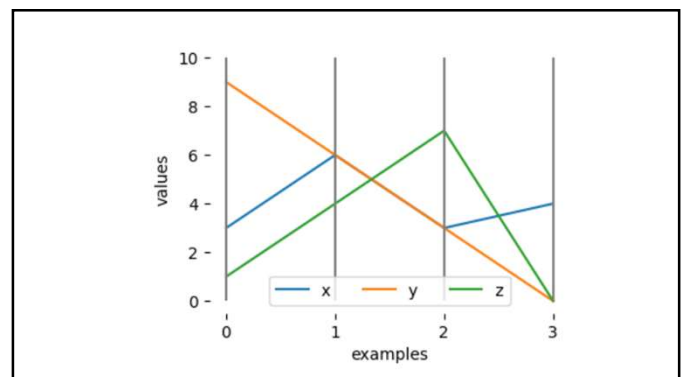
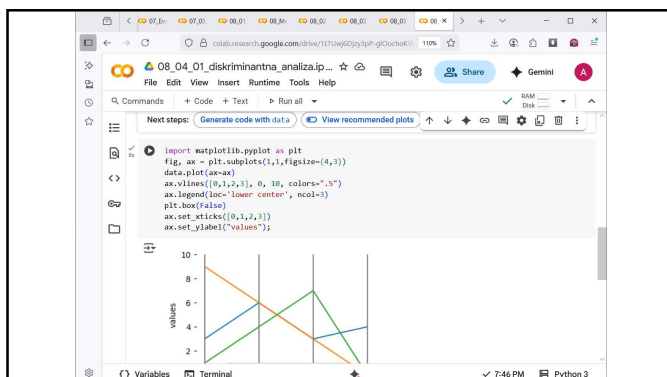
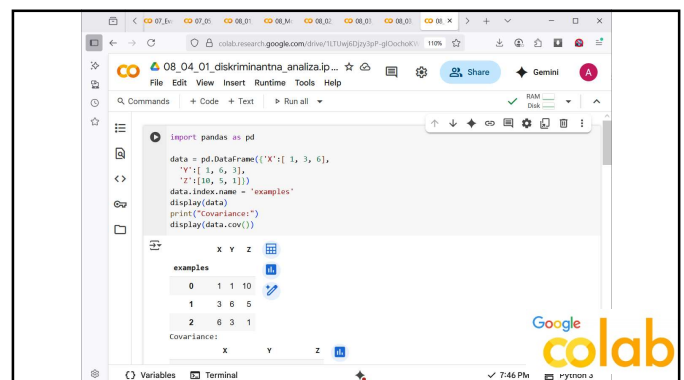
$$\text{Cov}(X, Y) = 66.67$$

- **Matrica kovarijanci** je kvadratna matrica koja opisuje koliko su sve značajke međusobno povezane (korelirane) u višedimenzionalnim podacima.

$$C = \begin{bmatrix} \text{Cov}(X_1, X_1) & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Cov}(X_2, X_2) & \dots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \dots & \text{Cov}(X_n, X_n) \end{bmatrix}$$

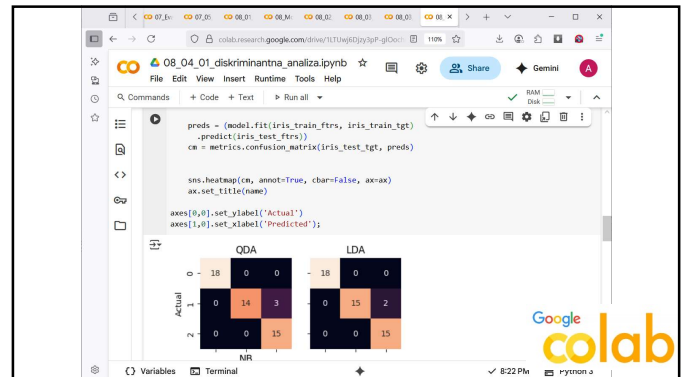
- Dijagonala sadrži varijance pojedinih značajki
- Ostali elementi su kovarijance između različitih značajki

Google
colab

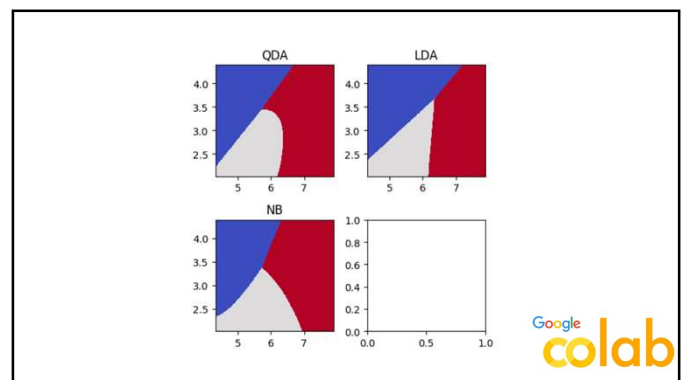
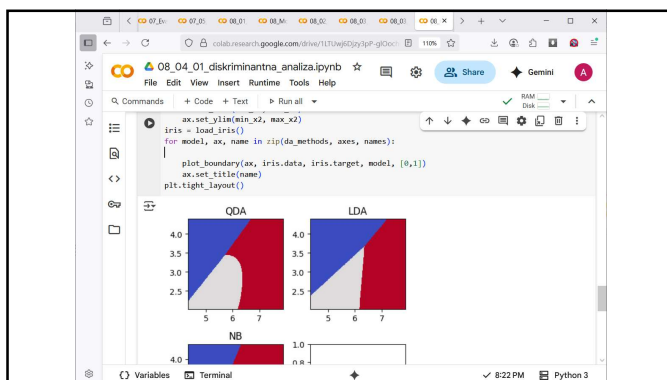


- 1. QDA: moguće različite matrice kovarijance po klasi
- 2. LDA: ista matrica kovarijance za sve klase,
- 3. GNB: različite dijagonalne matrice kovarijance po klasi
- 4. DLDA: ista dijagonalna matrica kovarijance za sve klase.

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colab



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8.5. Usporedba klasifikatora – digits dataset

